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(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Yash Khadgi
2)Rudrakshi Bhandekar
3)Vedant Nawghare
4)Dr. Priya Dasarwar

(57) Abstract :
ABSTRACT A STACKING ENSEMBLE MACHINE LEARNING SYSTEM AND METHOD FOR AGRICULTURAL CROP PRICE PREDICTION The present invention discloses a stacking ensemble machine learning system (100) and method for agricultural crop price prediction. The system comprises a data collection module (110) for acquiring multi-source agricultural data including soil nutrients, weather conditions, and historical prices; a preprocessing module (120) implementing missing value imputation, categorical encoding, feature scaling, and outlier treatment; a feature selection module (130) employing correlation analysis, VIF assessment, and importance scoring; a model training module (140) comprising XGBoost (141), Ridge Regression (142), and Random Forest (143) base learners; a stacking ensemble module (150) with an XGBoost meta-learner (151) that aggregates base model predictions to generate optimized price forecasts; and a prediction output module (160) delivering results to end users. The system achieves R-squared accuracy of 91.36% and MAE of 0.18, significantly outperforming individual models, providing farmers and policymakers with reliable decision support for agricultural market planning. [to be published with fig.1]

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